YiRage (YPK) — 系统架构 · RuntimeFusion 数据流

Inference engine owns scheduling/KV; RF owns fusion identity; search and execute use the same backend

Primary flow
Fallback / feedback
Dashed = architecture zone
Box = module + responsibility

PROJECT OVERVIEW YiRage is a multi-backend LLM kernel compiler and runtime centered on graph superoptimization and RuntimeFusion. It searches backend-specific fusion plans offline, then injects verified FusionCapsules into vLLM or SGLang decode steps.

INFERENCE ENGINE · 调度与 KV 所有者

- **vLLM** — continuous batch · PagedAttention
- **SGLang** — RadixAttention · extend meta
- **Decode Metadata** — block table · seq lens · SM budget

RUNTIMEFUSION · 每个 decode step

- **RF.step(inputs, meta)** — select 0..N capsules
- **FusionCapsule** — MLP gate_up + down segment pipeline
- **Native Fallback** — skip → engine native ops

SEARCH & PLAN · offline / warm-up

- **superoptimize(backend)** — KN / TB graph search
- **FusionPlan / μGraph** — profile · fingerprint verify
- **Persistent Cache** — shape bucket restore
- **Ray Coordinator** — optional distributed search — fallback / re-search

BACKEND MANAGER · 编译与执行边界

- **BackendRegistry** — search + execute same backend
- **KNGraph / TBGraph** — transpiler → kernels
- **yirage.core** — Cython runtime

BACKENDS → HARDWARE

- **CUDA / MACA** — NVIDIA GPU · MetaX C500/C550
- **ROCm / MPS** — AMD GPU · Apple Silicon
- **Ascend / NKI** — Huawei 910 · AWS Neuron
- **CPU / MLIR / Triton** — x86 · ARM · extra targets

主链：Engine metadata → RF.step → FusionCapsule → BackendRegistry → target hardware。